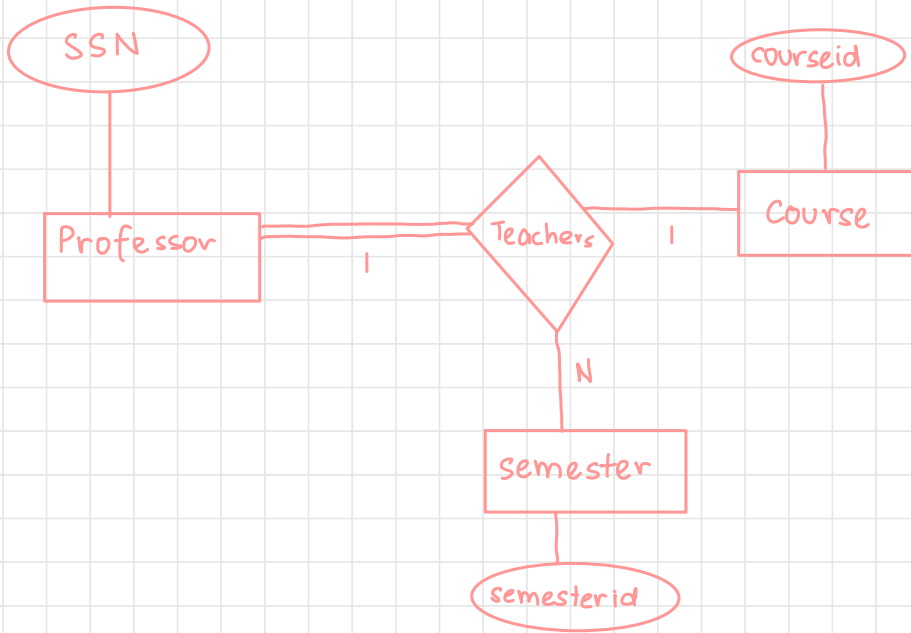


Homework 3

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Problem 1:



Problem 11

1. left only right only left-right buckets

L

L and R

R

A

BD

CE

$$\{A\}^+ = \{A\}$$

$$= \{A, C, D\} \quad A \rightarrow CD \Rightarrow A \rightarrow C$$

$$= \{A, C, D, B\} \quad D \rightarrow B$$

$$\{A\}^+ = \{A, C, D, B, E\} \quad B \rightarrow E$$

The highest normal form this design satisfies is 1NF.

Candidate key = {A}

Problem 11

1. $C \rightarrow D, C \rightarrow A, B \rightarrow C$

$$B^+ = \{B, C, D, A\}$$

Candidate key = $\{B\}$

$$\left\{ \begin{array}{ll} C \rightarrow D & 2NF \\ C \rightarrow A & 2NF \\ B \rightarrow C & BCNF \end{array} \right\} = 2NF$$

2. $B \rightarrow C, D \rightarrow A$

$$BD^+ = \{B, C, D, A\}$$

Candidate key = $\{BD\}$

$$\left\{ \begin{array}{ll} B \rightarrow C & 1NF \\ D \rightarrow A & 1NF \end{array} \right\} = 1NF$$

3. $ABC \rightarrow D, D \rightarrow A$

$$ABC^+ = \{A, B, C, D\}$$

$$DBC^+ = \{D, A, B, C\}$$

Candidate key = $\{ABC, BCD\}$

$$\left\{ \begin{array}{ll} ABC \rightarrow D & BCNF \\ D \rightarrow A & 3NF \end{array} \right\} = 3NF$$

4. $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

$$A^+ = \{A, B, C, D\}$$

Candidate key = $\{A\}$

$$\left\{ \begin{array}{ll} A \rightarrow B & BCNF \\ BC \rightarrow D & 2NF \\ A \rightarrow C & BCNF \end{array} \right\} = 2NF$$

5. $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

$$AB^+ = \{A, B, C, D\}$$

$$CD^+ = \{C, D, A, B\}$$

$$AD^+ = \{A, D, B, C\}$$

$$CB^+ = \{C, B, A, D\}$$

Candidate key = $\{AB, AD, CB, CD\}$

$$\left\{ \begin{array}{ll} AB \rightarrow C & BCNF \\ AB \rightarrow D & BCNF \\ C \rightarrow A & 3NF \\ D \rightarrow B & 3NF \end{array} \right\} = 3NF$$